

Recursion - Advanced

Tree recursion, tail recursion, and recursion patterns.

Key concepts

- Tree recursion: multiple calls per step (e.g. fibonacci).
- Tail recursion can be optimised to a loop.
- Watch stack depth to avoid overflow.

Key idea / example

`fib(n)=fib(n-1)+fib(n-2)` is exponential without memoisation.

Today's tasks (1.5h)

- ☐ Write fibonacci recursively
- ☐ Spot the repeated work
- ☐ Add memoisation

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)